THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** ampere	1 A = 1 C/s
2	**(c)** 2 A	$I = Q/t$
3	**(c)** l/A	$R = \rho l/A$
4	**(b)** Incorrect — it is connected in parallel	Ammeter in series
5	**(a)** Student A is correct	Add in series
6	**(c)** $H = I^2Rt$	Joule's law
7	**(b)** 3.6×10^6 J	1000 W × 3600 s
8	**(d)** 484 Ω	$R = V^2/P$
9	**(b)** tungsten	High melting point
10	**(d)** 10 V	$V = IR$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	At constant temperature, the current through a conductor is directly proportional to the potential difference across it: $V = IR$.	2
12	Potential difference when 1 J of work is done to move 1 C of charge: 1 V = 1 J/C.	2
13	High resistivity and they do not oxidise (burn) readily at high temperatures.	2
14	False — they are connected in parallel so each gets the full voltage and can be switched independently.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{1}{R} = \frac{1}{4} + \frac{1}{6} + \frac{1}{12} = \frac{6}{12} \Rightarrow R = 2\ \Omega$; $I = 12/2 = 6$ A	2+1
16	Energy = 1 × 2 × 30 = 60 kWh = 60 units; cost = 60 × 6 = ₹360	2+1
17	$H = I^2Rt = 4 \times 50 \times 10 = 2000$ J	3
	OR: Resistance of a conductor of unit length and unit cross-section (Ω m); depends on length, area of cross-section, nature of material (and temperature)	1+2
18	Cell, key, resistor and ammeter in series; voltmeter in parallel across the resistor	2+1
	OR: Length × 2, area ÷ 2 ⇒ $R' = 4R = 40\ \Omega$	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $V = V_1 + V_2 + V_3 = I(R_1 + R_2 + R_3) \Rightarrow R_s = R_1 + R_2 + R_3$ (ii) $R = 15\ \Omega$, $I = 0.4$ A, $V_1 = 2$ V, $V_2 = 4$ V	2+3
	OR: (i) $I = I_1 + I_2 + I_3$, $\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} \Rightarrow \frac{1}{R_p} = \sum \frac{1}{R_i}$ (ii) each device gets the full voltage; one failing does not stop others; lower total resistance	3+2
20	(i) 4×60×5 = 1.2 kWh; 1×2 = 2 kWh; total 3.2 kWh (ii) 3.2×30×5 = ₹480 (iii) $I = 1000/220 = 4.55$ A (iv) melts on excess current to protect circuit	2+1+1+1
	OR: (i) $H = I^2Rt$ (ii) electric iron, toaster, bulb, fuse (iii) $P = VI = 1100$ W; $E = 1.1 \times 2 = 2.2$ kWh	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME — SET B

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** ohm metre	Ω m
2	**(a)** 6.25×10^{18}	$1/(1.6 \times 10^{-19})$
3	**(b)** 3 Ω	$R/2$
4	**(a)** Correct	Resistivity
5	**(b)** Student A is correct	$P = VI$
6	**(d)** voltmeter	Parallel
7	**(d)** resistance	V/I
8	**(b)** 24 J	$W = QV$
9	**(d)** kilowatt hour	1 unit
10	**(a)** 0.27 A	$I = P/V$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rate of flow of electric charge, $I = Q/t$; ampere.	1+1
12	Very high melting point (3380 °C) and it emits light at high temperature.	2
13	$I = V/R = 12/40 = 0.3$ A	2
14	False — resistance is inversely proportional to the area of cross-section.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Parallel: $\frac{6\times3}{9} = 2$ Ω; total = 6 Ω; $I = 12/6 = 2$ A	2+1
16	(i) $R = V^2/P = 484$ Ω (ii) $I = P/V = 0.45$ A (iii) $E = 0.1 \times 10 = 1$ kWh	1+1+1
17	The cord is made of copper with low resistance, so little heat is produced; the element (nichrome) has high resistance and produces more heat ($H = I^2Rt$)	3
	OR: $R = \rho l/A = 1.6 \times 10^{-8} \times 1/10^{-6} = 0.016$ Ω	3
18	Rate of consumption of electrical energy; $P = VI = I^2R = V^2/R$; watt	1+1+1
	OR: $R = V/I = 44$ Ω; $176/n = 44 \Rightarrow n = 4$	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) cell, key, ammeter, resistor (nichrome wire), rheostat in series and voltmeter across resistor; record V and I for different settings; V/I is constant (ii) straight line through origin (iii) resistance	3+1+1
	OR: (i) same current; one fails — all stop; voltage divides (ii) $\frac{1}{R} = \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2} \Rightarrow R = 2$ Ω; $I = 4$ A; currents 2 A, 1 A, 1 A	2+3
20	(i) series (ii) parallel (iii) series: $I = 10/20 = 0.5$ A; parallel: total $R = 5$ Ω, $I = 2$ A (1 A in each) (iv) series: other goes off; parallel: other keeps glowing	1+1+2+1
	OR: (i) $W = QV = (It)(IR) = I^2Rt$ (ii) $H = 16 \times 50 \times 60 = 48\,000$ J	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.